

Total No. of Questions : 8]

SEAT No. :

P647

[Total No. of Pages : 2

[5869]-274

S.E. (Computer Engineering)

COMPUTER GRAPHICS

(2019 Pattern) (Semester - III) (210244)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.No.6 and Q.No.7 or Q.No.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) Consider a polygon with 4 sides as follow $P_1(1, 1)$, $P_2(3, 1)$, $P_3(3, 3)$, $P_4(1, 3)$. Scale this polygon to half of its size. Draw initial and scaled polygon. [6]
- b) What is parallel projection? Explain in brief about each type of parallel projections? [6]
- c) Explain rotation about an arbitrary point in 2D. [6]

OR

- Q2)** a) Consider a square A(1, 0), B(0, 0), C(0, 1) and D(1, 1). Show the steps to rotate the given square by 45° clockwise about the point A(1, 0). Draw initial and rotated square. [6]
- b) What is perspective projection? Explain in brief about each type of perspective projections? [6]
- c) Explain rotation about an arbitrary axis in 3D. [6]

- Q3)** a) Explain and compare point source and diffuse illumination. [6]
- b) Explain Z-buffer (Depth Buffer) algorithm with example. [6]
- c) List properties of light. [5]

OR

- Q4)** a) Explain and compare diffuse and specular reflection. [6]
- b) Explain Painters (Depth Sort) algorithm with example. [6]
- c) What is the need of hidden surface algorithm? [5]

P.T.O.

- Q5)** a) What is interpolation? Write short note on interpolating algorithm. [6]
b) Explain Hilbert curve and give its fractal dimension. [6]
c) How is coastline measured? What are the methods of measuring length? [6]

OR

- Q6)** a) Derive blending function of Bezier curve. How blending function is calculated for cubic polynomial curve? [6]
b) Explain Triadic curve and give its fractal dimension. [6]
c) How do you find the area under a normal curve? [6]
- Q7)** a) Write short note on motion specification. [6]
b) Draw block diagram of NVIDIA workstation and explain it in brief. [6]
c) Write algorithms to create and delete a segment. [5]

OR

- Q8)** a) Explain following terms with example : [6]
i) Key frames.
ii) Morphing.
b) Explain Intel i860 processor with block diagram. [6]
c) What does GPU stand for? What is the role of GPU in gaming? List any four GPUs used in gaming. [5]

